Chapter 2. Probability, Likelihood and Information Theory

Random variables, maximum likelihood, MAP estimation, Bayesian models; Entropy, KL divergence, mutual information, cross-entropy

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1 Events and Conditional Probability

A sample space Ω contains the possible outcomes and events are (measurable) subsets. The axioms are nonnegativity, $P(\Omega) = 1$ and countable additivity for disjoint events. They imply

$$P(\emptyset) = 0, \qquad P(E^c) = 1 - P(E), \qquad P(E \cup F) = P(E) + P(F) - P(E \cap F),$$

monotonicity $E \subseteq F \Rightarrow P(E) \le P(F)$, and continuity along increasing or decreasing sequences of events.

Definition 2.1: conditional probability, Bayes' rule, independence

For $P(E) > 0$, $P(F \mid E) = P(F \cap E)/P(E)$. For a partition $E_1, E_2, \dots$ of Ω,

$$P(F) = \sum_i P(F \mid E_i)P(E_i), \qquad P(E_i \mid F) = \frac{P(F \mid E_i)P(E_i)}{\sum_j P(F \mid E_j)P(E_j)}.$$

E, F are independent if $P(E \cap F) = P(E)P(F)$.

Example 2.2: why the prior matters

Let $P(D) = 0.01$, $P(+ \mid D) = 0.9$, $P(+ \mid D^c) = 0.1$. Then

$$P(D \mid +) = \frac{0.9(0.01)}{0.9(0.01) + 0.1(0.99)} \approx 0.0833.$$

A sensitive test does not make the posterior large when the prior is small. Two fair coin tosses: "first head" and "second head" each have probability $1/2$ and their intersection $1/4$, so they are independent.

2 Random Variables and Distributions

Motivation. Learning treats data as samples from an unknown distribution, and losses are expectations. We fix the notation for distributions, expectations and independence used in every later chapter.

A random variable is a measurable function $X : \Omega \to \mathbb{R}$ (e.g. the number of heads in three tosses, $X(HHT) = 2$). Its CDF is $F_X(t) = P(X \le t)$.

- Discrete: probability mass $p(x) = P(X = x)$, $F_X(t) = \sum_{x \le t} p(x)$.
- Continuous: density $f \ge 0$ with $\int f = 1$, $P(a < X \le b) = \int_a^b f(u)\,du$. A density value is not a probability and may exceed one; $P(X = x) = 0$.

$$\mathbb{E}[g(X)] = \sum_x g(x)p(x) \text{ or } \int g(x)f(x)\,dx, \qquad \operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

Theorem 2.3: tools used throughout the course

(i) Linearity: $\mathbb{E}[aX + bY] = a\mathbb{E}X + b\mathbb{E}Y$ always.

(ii) Independence: $\mathbb{E}[XY] = \mathbb{E}X\,\mathbb{E}Y$; hence $\operatorname{Var}(X+Y) = \operatorname{Var}X + \operatorname{Var}Y$. Uncorrelatedness ($\operatorname{Cov} = 0$) suffices for the variance identity.

(iii) Tower property: $\mathbb{E}[\mathbb{E}[Z \mid X]] = \mathbb{E}Z$.

(iv) $\mathbb{E}[(Z-c)^2] = \operatorname{Var} Z + (\mathbb{E}Z - c)^2 \geq \operatorname{Var} Z$ for every constant c.

Proof. (iv): write $Z - c = (Z - \mathbb{E}Z) + (\mathbb{E}Z - c)$, square, and use $\mathbb{E}[Z - \mathbb{E}Z] = 0$ to kill the cross term. The other items are standard. ■

3 Maximum Likelihood Estimation

Definition 2.4: likelihood and MLE

For iid data $x_1, \ldots, x_n$ from $p(x \mid \theta)$,

$$L(\theta) = \prod_{i=1}^{n} p(x_i \mid \theta), \qquad \hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} \sum_{i=1}^{n} \log p(x_i \mid \theta).$$

The data are fixed and the parameter varies; L is not a distribution over θ. The logarithm preserves the maximizer and turns products into sums.

Example 2.5: Bernoulli, including the boundary

$x_i \in \{0, 1\}$ with $P(x_i = 1) = \theta$; let $s = \sum x_i$, $f = n - s$. Then $\log L = s \log \theta + f \log(1 - \theta)$. For $0 < s < n$,

$$\frac{d}{d\theta} \log L = \frac{s}{\theta} - \frac{f}{1-\theta} = 0 \iff \theta = \frac{s}{n}, \qquad \frac{d^2}{d\theta^2} \log L < 0.$$

If $s = 0$, $L = (1-\theta)^n$ is decreasing and the maximizer is 0; if $s = n$ it is 1. Both agree with s/n, but the derivative argument cannot be used there. Eight heads in ten tosses gives 0.8; two positive reviews out of two gives the extreme estimate 1.

Example 2.6: Gaussian

For $x_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, $\log L = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum (x_i - \mu)^2$, giving $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$ (biased; the unbiased version divides by $n - 1$).

4 Bayesian Inference and MAP

Motivation. Maximum likelihood can overfit with little data. Bayesian inference adds a prior and returns a posterior; its mode (MAP) is a single estimate that will turn out to be maximum likelihood plus a regularizer.

Definition 2.7: prior, posterior, evidence, MAP

With prior $p(\theta)$ and likelihood $p(D \mid \theta)$,

$$p(\theta \mid D) = \frac{p(D \mid \theta) p(\theta)}{p(D)}, \qquad p(D) = \int p(D \mid \theta) p(\theta)\, d\theta,$$

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} \{\log p(D \mid \theta) + \log p(\theta)\}.$$

The evidence does not depend on θ. MAP is the posterior mode; the posterior mean is a different summary.

4.1 Beta–Bernoulli conjugacy

Theorem 2.8: Beta prior

If $\theta \sim \mathrm{Beta}(a, b)$, i.e. $p(\theta) = \theta^{a-1}(1-\theta)^{b-1}/B(a,b)$ with $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, and s successes, f failures are observed, then

$$\theta \mid D \sim \mathrm{Beta}(a+s,\, b+f), \qquad \mathbb{E}[\theta \mid D] = \frac{a+s}{a+b+n}, \qquad \hat{\theta}_{\mathrm{MAP}} = \frac{a+s-1}{a+b+n-2}$$

(the mode formula needs both posterior parameters above one).

Proof. $p(\theta \mid D) \propto \theta^s(1-\theta)^f \cdot \theta^{a-1}(1-\theta)^{b-1} = \theta^{(a+s)-1}(1-\theta)^{(b+f)-1}$, which is the $\mathrm{Beta}(a+s, b+f)$ kernel. For $\mathrm{Beta}(\alpha,\beta)$, $\mathbb{E}\theta = B(\alpha+1,\beta)/B(\alpha,\beta) = \alpha/(\alpha+\beta)$ using $\Gamma(z+1) = z\Gamma(z)$. The mode solves $\frac{\alpha-1}{\theta} = \frac{\beta-1}{1-\theta}$, i.e. $\theta = \frac{\alpha-1}{\alpha+\beta-2}$. ∎

The uniform prior is $\mathrm{Beta}(1,1)$; then the posterior mean $(s+1)/(n+2)$ is Laplace's rule of succession, and it equals the posterior predictive probability of a success, since $P(X_{\mathrm{new}} = 1 \mid D) = \int \theta\, p(\theta \mid D)\, d\theta$.

Example 2.9: the two sellers

Seller 1 has 90 positive and 10 negative reviews; seller 2 has 2 positive and 0 negative. With uniform priors

$$\theta_1 \mid D_1 \sim \mathrm{Beta}(91, 11), \qquad \theta_2 \mid D_2 \sim \mathrm{Beta}(3, 1).$$

Posterior predictive probabilities of a positive next review: $91/102 \approx 0.892$ versus $3/4 = 0.75$, so seller 1 is preferred, whereas the MLE (0.9 versus 1) prefers seller 2. A different question is which unknown rate is larger: since $\mathrm{Beta}(3,1)$ has CDF t^3,

$$P(\theta_1 > \theta_2 \mid D) = \mathbb{E}[\theta_1^3] = \frac{91 \cdot 92 \cdot 93}{102 \cdot 103 \cdot 104} \approx 0.713.$$

5 MAP as Regularization

Motivation. A Gaussian prior turns MAP into ℓ_2-penalized likelihood, so regularization has a probabilistic meaning.

5.1 Gaussian mean with a Gaussian prior

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$ and prior $\theta \sim N(0, \tau^2)$.

Theorem 2.10: MAP is ridge, and it shrinks

$$\hat{\theta}_{\mathrm{MAP}} = \arg\min_{\theta} \sum_{i=1}^{n} (X_i - \theta)^2 + \lambda\theta^2, \quad \lambda = \frac{1}{\tau^2}, \qquad \hat{\theta}_{\mathrm{MAP}} = a\bar{X}, \quad a = \frac{n}{n+\lambda} = \frac{n\tau^2}{n\tau^2+1}.$$

Proof. $-\log p(\theta \mid X) = \frac{1}{2}\sum(X_i-\theta)^2 + \frac{\theta^2}{2\tau^2} + \text{const}$; multiplying by 2 gives the ridge objective. Setting the derivative $-2\sum(X_i-\theta)+2\lambda\theta$ to zero gives $(n+\lambda)\theta = n\bar{X}$. ■

Remark 2.11: Laplace prior gives the lasso

With $p(\theta) \propto e^{-|\theta|/b}$ instead, $-\log$ posterior is $\frac{1}{2}\sum(X_i-\theta)^2 + \frac{1}{b}|\theta|$ and the MAP is the soft threshold $\operatorname{sign}(\bar{X})\max(|\bar{X}| - \frac{1}{nb}, 0)$ (Chapter 3).

6 Information and Entropy

Motivation. Classification losses are built from information theory. Entropy measures the average surprise of a distribution, and it is the baseline from which cross-entropy and KL divergence measure the cost of using the wrong distribution.

The information of an outcome is $h(x) = -\log p(x)$: rare outcomes carry more information, and for independent events probabilities multiply while information adds.

Definition 2.12: entropy

For a discrete random variable with pmf p,

$$H(X) = -\sum_x p(x)\log p(x) = \mathbb{E}[-\log p(X)], \qquad 0\log 0 := 0.$$

Natural logs give nats; base 2 gives bits.

Theorem 2.13: range of entropy

If X takes at most K values, $0 \le H(X) \le \log K$. $H = 0$ iff X is deterministic; $H = \log K$ iff X is uniform.

Proof. Each term $-p\log p \ge 0$ for $p \in [0,1]$, with equality iff $p \in \{0,1\}$. For the upper bound apply Jensen to the concave log: $H = \mathbb{E}\log\frac{1}{p(X)} \le \log \mathbb{E}\frac{1}{p(X)} = \log\sum_{x:p(x)>0} 1 \le \log K$, with equality iff $1/p(X)$ is constant on the support and the support has K points. ■

(For continuous densities the differential entropy can be negative; the statement above is for discrete variables.)

Definition 2.14: joint and conditional entropy

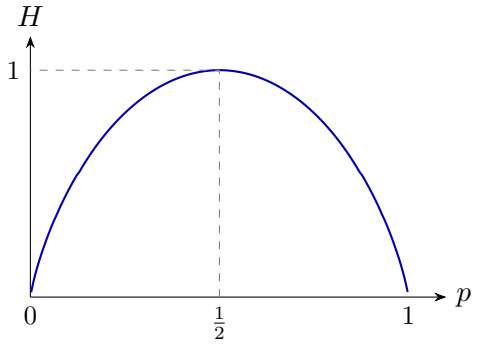
$$H(X,Y) = -\sum_{x,y} p(x,y)\log p(x,y),$$

$$H(Y \mid X) = -\sum_{x,y} p(x,y)\log p(y \mid x) = \sum_x p(x) H(Y \mid X = x).$$

Theorem 2.15: chain rule

$H(X,Y) = H(X) + H(Y \mid X)$.

Proof. Take $-\mathbb{E}$ of $\log p(X,Y) = \log p(X) + \log p(Y \mid X)$. ■



Entropy (in bits) of a Bernoulli(p) variable: zero for a certain outcome, maximal at $p = \frac{1}{2}$.

7 Kullback–Leibler Divergence

Motivation. How different are two distributions? KL divergence is the expected extra log-loss from coding with q instead of p. It is nonnegative and zero only at $q = p$, which is why minimizing it fits a model to data.

Definition 2.16: KL divergence

$$D_{\mathrm{KL}}(p \,\|\, q) = \sum_x p(x) \log \frac{p(x)}{q(x)} = \mathbb{E}_p\Big[\log \frac{p(X)}{q(X)}\Big].$$

Terms with $p(x) = 0$ contribute 0; if $p(x) > 0 = q(x)$ for some x the divergence is $+\infty$.

Theorem 2.17: Gibbs' inequality

$D_{\mathrm{KL}}(p \,\|\, q) \geq 0$, with equality iff $p = q$.

Proof. Sum over $S = \{x : p(x) > 0\}$. By Jensen for the concave log,

$$-D_{\mathrm{KL}}(p \,\|\, q) = \sum_{x \in S} p(x) \log \frac{q(x)}{p(x)} \leq \log \sum_{x \in S} p(x) \frac{q(x)}{p(x)} = \log \sum_{x \in S} q(x) \leq \log 1 = 0.$$

Equality forces (i) $q(x)/p(x) = c$ constant on S (Jensen) and (ii) $\sum_S q = 1$; then $c = \sum_S q = 1$, so $q = p$. Conversely $p = q$ gives $\log 1 = 0$ term by term. ■

Example 2.18: KL is not symmetric

For Bernoulli parameters, $D(r \,\|\, s) = r \log \frac{r}{s} + (1-r) \log \frac{1-r}{1-s}$. With $r = 0.9$, $s = 0.5$: $D(r\|s) \approx 0.368$ but $D(s\|r) \approx 0.511$ nats. KL is not a metric (no symmetry, no triangle inequality).

8 Mutual Information

Motivation. Mutual information is the KL divergence between a joint distribution and the product of its marginals: how much knowing X reduces uncertainty about Y. It is zero exactly under independence.

Definition 2.19: mutual information

$$I(X;Y) = D_{\mathrm{KL}}(p_{XY} \,\|\, p_X p_Y) = \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p(x)p(y)}.$$

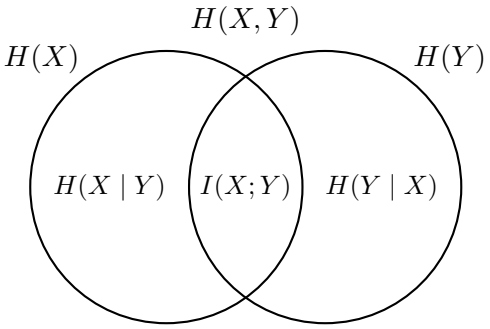
Theorem 2.20: identities

$$I(X;Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X) = H(X) + H(Y) - H(X,Y) \;\geq 0,$$

with $I = 0$ iff $X \perp Y$. Consequently $H(X \mid Y) \leq H(X)$ (conditioning reduces entropy).

Proof. $\log \frac{p(x,y)}{p(x)p(y)} = \log p(x \mid y) - \log p(x)$; take $\mathbb{E}$ and use $\sum_y p(x,y) = p(x)$ to get $-H(X \mid Y) + H(X)$. The last form follows from the chain rule. Nonnegativity and the equality case are Gibbs' inequality. ■

The entropy (Venn) diagram encodes $H(X,Y) = H(X \mid Y) + I(X;Y) + H(Y \mid X)$. For a fair bit X and $Y = X$, $I = 1$ bit; for independent fair bits, $I = 0$.



Entropy, joint and conditional entropy, and mutual information: the union of the two disks is $H(X,Y)$ and the overlap is $I(X;Y)$.

9 Cross-Entropy

Motivation. Cross-entropy is entropy plus KL divergence, so minimizing it over the model is the same as minimizing KL. This is why the standard classification loss is cross-entropy, and why it equals negative log-likelihood.

Definition 2.21: cross-entropy

$H(p,q) = -\sum_x p(x) \log q(x) = \mathbb{E}_p[-\log q(X)]$.

Theorem 2.22: cross-entropy = entropy + KL

$H(p,q) = H(p) + D_{\mathrm{KL}}(p \,\|\, q)$. Hence for fixed p, $\arg\min_q H(p,q) = \arg\min_q D_{\mathrm{KL}}(p \,\|\, q) = p$.

Proof. $-\sum p \log q = -\sum p \log p + \sum p \log(p/q)$. The second statement uses Gibbs. ■

Remark 2.23: classification loss and soft labels

For a one-hot label $\mathbf{e}_y$, $H(\mathbf{e}_y) = 0$ and $H(\mathbf{e}_y, q) = D_{\mathrm{KL}}(\mathbf{e}_y \| q) = -\log q_y$: the classification loss. $H(p, q)$ is *linear in its first argument*, $H(\alpha p + \beta p', q) = \alpha H(p, q) + \beta H(p', q)$, because p enters only as coefficients. Therefore a soft label that averages n one-hot labels gives $H(\frac{1}{n}\sum_j \mathbf{e}_{y_j}, q) = \frac{1}{n}\sum_j (-\log q_{y_j})$: fitting the label distribution by KL is equivalent (up to the constant $H(\hat{p})$) to ordinary cross-entropy on every individual label.